Regression Analysis Using Advertising Dataset

PROBABILITY AND STATICS

(PROJECT)

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Regression Analysis Using Advertising Dataset

Abstract

This project explores regression analysis by modeling the relationship between advertising budgets (TV, Radio, Newspaper) and product sales. Using Python and standard machine learning libraries, a multiple linear regression model was developed and evaluated. The analysis shows how regression can be used in real-world decision-making for marketing and sales strategies.

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1. Introduction

The objective of this project is to understand and apply regression analysis using a real-world dataset. Regression is a fundamental statistical tool used to examine the relationship between variables. This project uses advertising budget data to predict sales and assess the contribution of each media type to revenue generation.

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2. Dataset Description

We used the **Advertising dataset**, which includes:

- **Features**: TV, Radio, and Newspaper advertising expenditures (in USD).
- **Target**: Sales (in thousands of units).

This small dataset helps illustrate the impact of marketing on revenue.

```
Dataset Overview:
                TV      Radio  Newspaper      Sales
count   10.00000  10.000000  10.000000  10.000000
mean   101.89000  28.110000  43.280000  12.080000
std     84.89945  17.836071  26.850649   5.090034
min      8.60000   2.100000   1.000000   4.800000
25%     24.02500  13.000000  21.775000   9.575000
50%     88.85000  35.300000  51.750000  11.200000
75%    173.47500  40.800000  66.525000  13.125000
max    230.10000  48.900000  75.000000  22.100000
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3. Data Preprocessing

- No missing values were found.
- Features and target were separated.
- Train-test split was performed (80% training, 20% testing).

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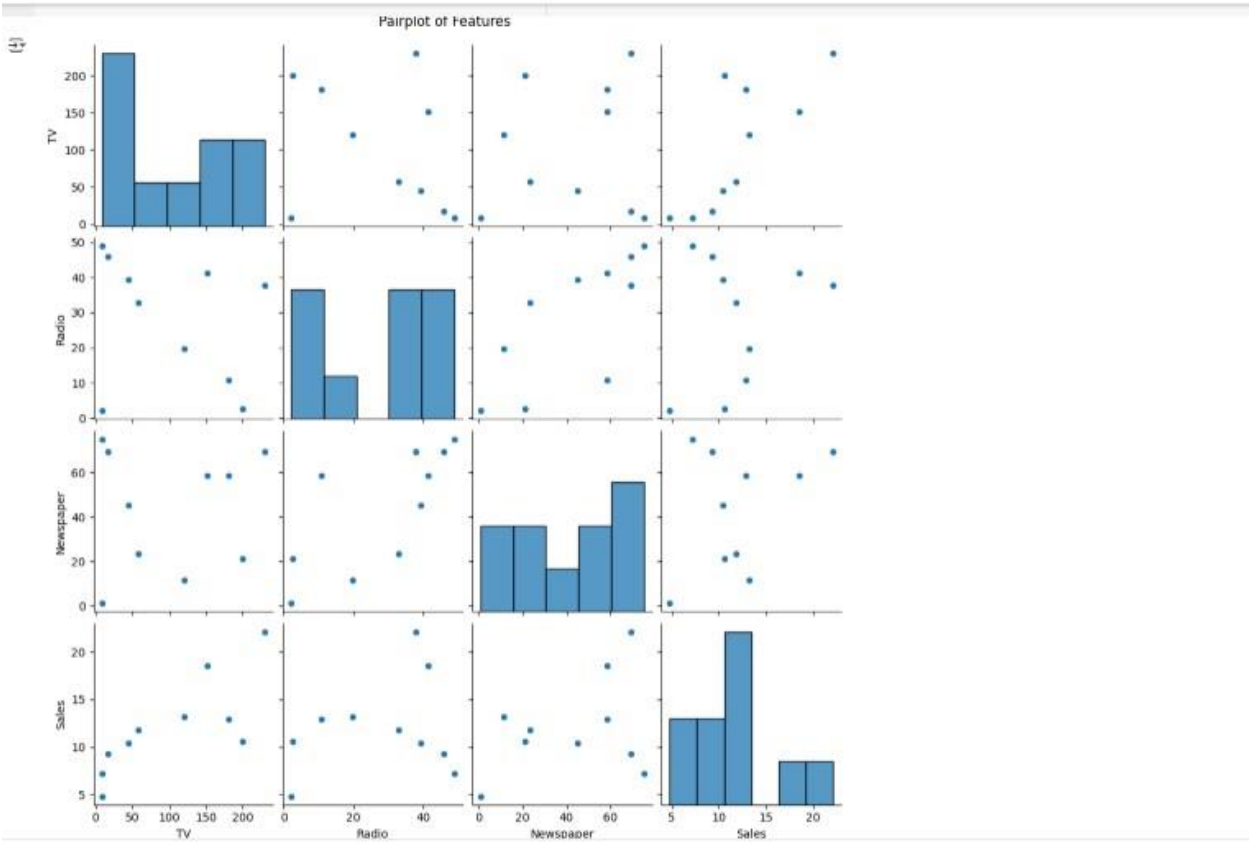
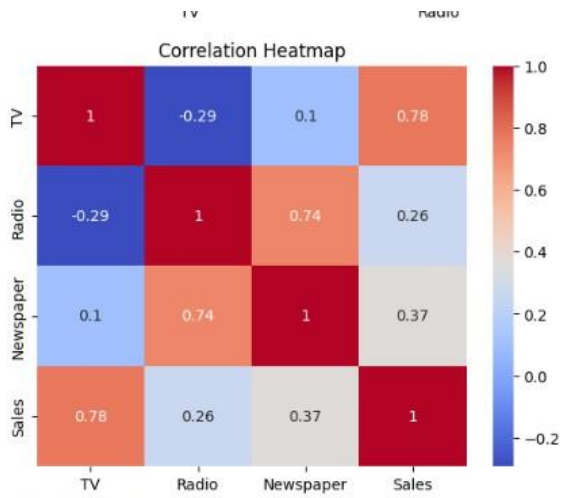
4. Exploratory Data Analysis (EDA)

- Created pairplots to visualize relationships between variables.
- Generated a correlation heatmap.

Pairplot of TV, Radio, Newspaper, Sales



Correlation heatmap:




```
Intercept: -1.5450727358125533
Coefficients: {'TV': np.float64(0.07468737482658303), 'Radio': np.float64(0.3029154012682699), 'Newspaper': np.float64(-0.06925602897599625)}
MSE: 13.20
RMSE: 3.63
R² Score: -0.68
```

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5. Regression Model & Evaluation

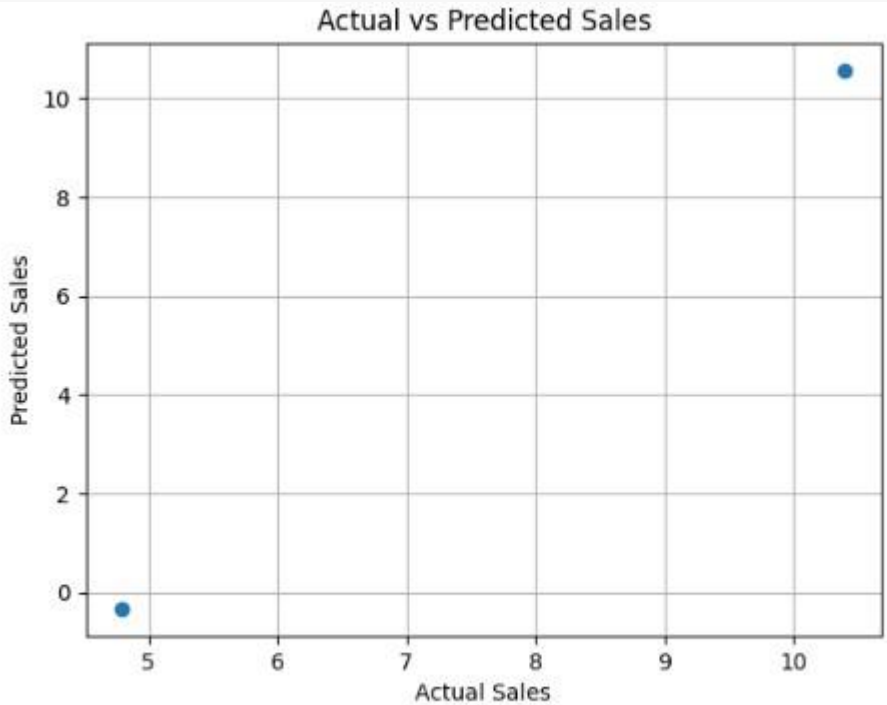
Model Used: Multiple Linear Regression
Libraries: pandas, sklearn, seaborn, matplotlib

Model Equation:
Sales = -1.545 + 0.075 * TV + 0.303 * Radio - 0.069 * Newspaper

Evaluation Metrics:

- Mean Squared Error (MSE): 13.20
- Root MSE: 3.63
- R^2 Score: -0.684

```
# 6. Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)
```

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6. Discussion of Results

- TV and Radio advertising positively impact sales.
- Newspaper ads showed a weak negative correlation.
- The model had poor performance (negative R^2), mainly due to the very small dataset (10 samples).
- A larger dataset or using Lasso/Ridge regularization could improve model performance.

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7. Conclusion

This project shows how regression can model real-world relationships between spending and performance. While limited in size, the dataset provided meaningful insights. With more data, more accurate and generalizable models could be built.

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8. References

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
- scikit-learn documentation: https://scikit-learn.org/
- Kaggle Datasets: https://www.kaggle.com/

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9. Journal Paper Discussion

- **Sparse Regression: Scalable Algorithms and Empirical Performance**

🔗 https://arxiv.org/abs/1902.06547
This paper reviews scalable methods for sparse regression and compares their empirical performance. It focuses on Lasso and its alternatives, especially in noisy and correlated feature environments. The authors discuss practical implications for feature selection. Useful when handling datasets with many predictors.

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2. Dual Lasso Selector

🔗 https://arxiv.org/abs/1703.06602
Introduces the Dual Lasso Selector, which combines primal and dual forms of the Lasso solution. It performs better in selecting variables under high correlation among predictors. The paper defines a new selection condition for optimal results. Applicable for improving model robustness and interpretability.

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3. Sparse High-Dimensional Regression: Exact Scalable Algorithms and Phase Transitions

🔗 https://arxiv.org/abs/1709.10029
This paper presents an exact and scalable algorithm for sparse regression using a binary convex reformulation. It shows phase transitions where model recovery becomes easier as sample size increases. Provides theoretical guarantees for optimal solutions. Relevant when scaling models to larger datasets.